www.tntextbooks.in

The equation for the x coordinate of the center of mass is,

$$x_{CM} = \frac{\sum m_i x_i}{\sum m_i}$$

where, $\sum m_i$ is the total mass M of all the particles, $\left(\sum m_i = M\right)$. Hence,

$$x_{CM} = \frac{\sum m_i x_i}{M} \qquad (5.1)$$

Similarly, we can also find y and z coordinates of the center of mass for these distributed point masses as indicated in Figure (5.2).

$$y_{CM} = \frac{\sum m_i y_i}{M} \qquad (5.2)$$

$$z_{CM} = \frac{\sum m_i z_i}{M} \qquad (5.3)$$

Hence, the position of center of mass of these point masses in a Cartesian coordinate system is (x_{CM}, y_{CM}, z_{CM}). In general, the position of center of mass can be written in a vector form as,

$$\vec{r}_{CM} = \frac{\sum m_i \vec{r}_i}{M} \qquad (5.4)$$

where, $\vec{r}_{CM} = x_{CM}\hat{i} + y_{CM}\hat{j} + z_{CM}\hat{k}$ is the position vector of the center of mass and $\vec{r}_i = x_i\hat{i} + y_i\hat{j} + z_i\hat{k}$ is the position vector of the distributed point mass; where, $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vectors along X, Y and Z-axes respectively.

5.1.4 Center of Mass of Two Point Masses

With the equations for center of mass, let us find the center of mass of two point masses m_1 and m_2, which are at positions x_1 and x_2 respectively on the X-axis. For this case, we can express the position of center of mass in the following three ways based on the choice of the coordinate system.

(i) When the masses are on positive X-axis:
The origin is taken arbitrarily so that the masses m_1 and m_2 are at positions x_1 and x_2 on the positive X-axis as shown in Figure 5.3(a). The center of mass will also be on the positive X-axis at x_{CM} as given by the equation,

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

(ii) When the origin coincides with any one of the masses:
The calculation could be minimised if the origin of the coordinate system is made to coincide with any one of the masses as shown in Figure 5.3(b). When the origin coincides with the point mass m_1, its position x_1 is zero, (i.e. $x_1 = 0$). Then,

$$x_{CM} = \frac{m_1(0) + m_2 x_2}{m_1 + m_2}$$

The equation further simplifies as,

$$x_{CM} = \frac{m_2 x_2}{m_1 + m_2}$$

(iii) When the origin coincides with the center of mass itself:
If the origin of the coordinate system is made to coincide with the center of mass,